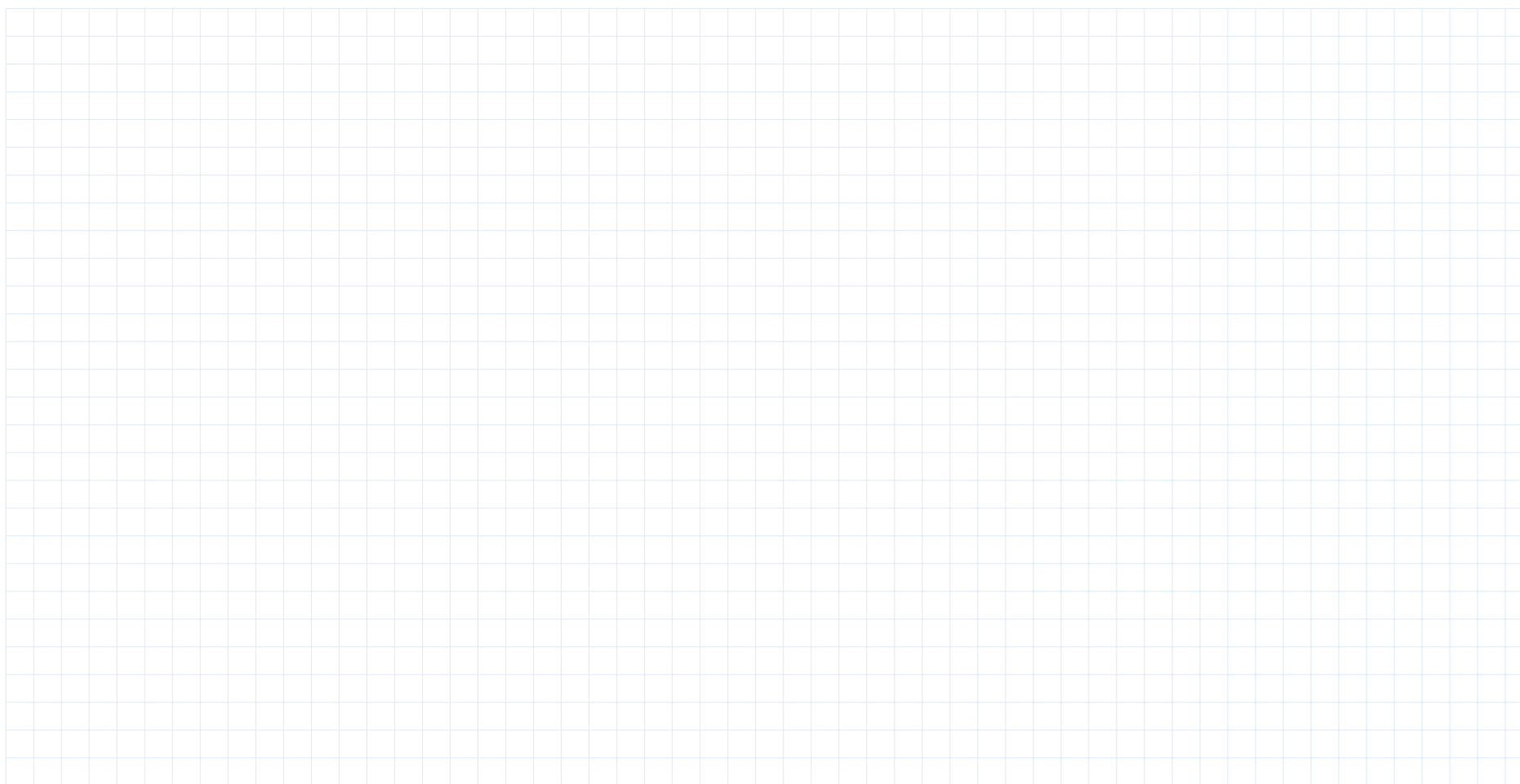


$$\sin(2\theta) = 2 \sin \theta \cos \theta, \quad \cos(2\theta) = \cos^2 \theta - \sin^2 \theta.$$

If $\sin \theta = \frac{3}{5}$ and $\frac{\pi}{2} < \theta < \pi$, find $\sin(2\theta)$ and $\cos(2\theta)$ exactly.



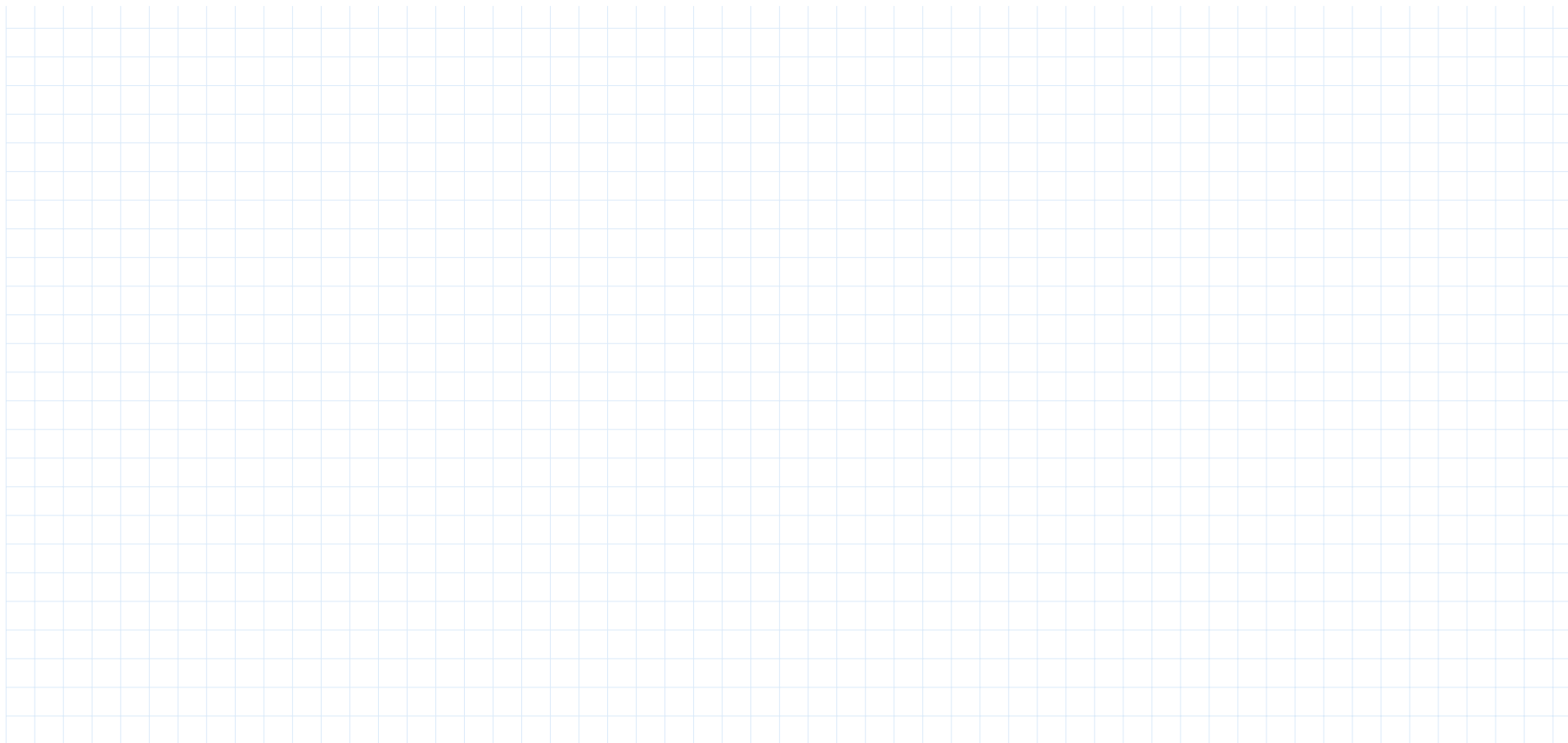
Derive a Tangent Formula

The cosine double-angle formula can also be written as

$$\cos(2\theta) = 1 - 2\sin^2 \theta = 2\cos^2 \theta - 1.$$

Use the sine and cosine double-angle formulas to derive

$$\tan(2\theta) = \frac{2\tan \theta}{1 - \tan^2 \theta}.$$

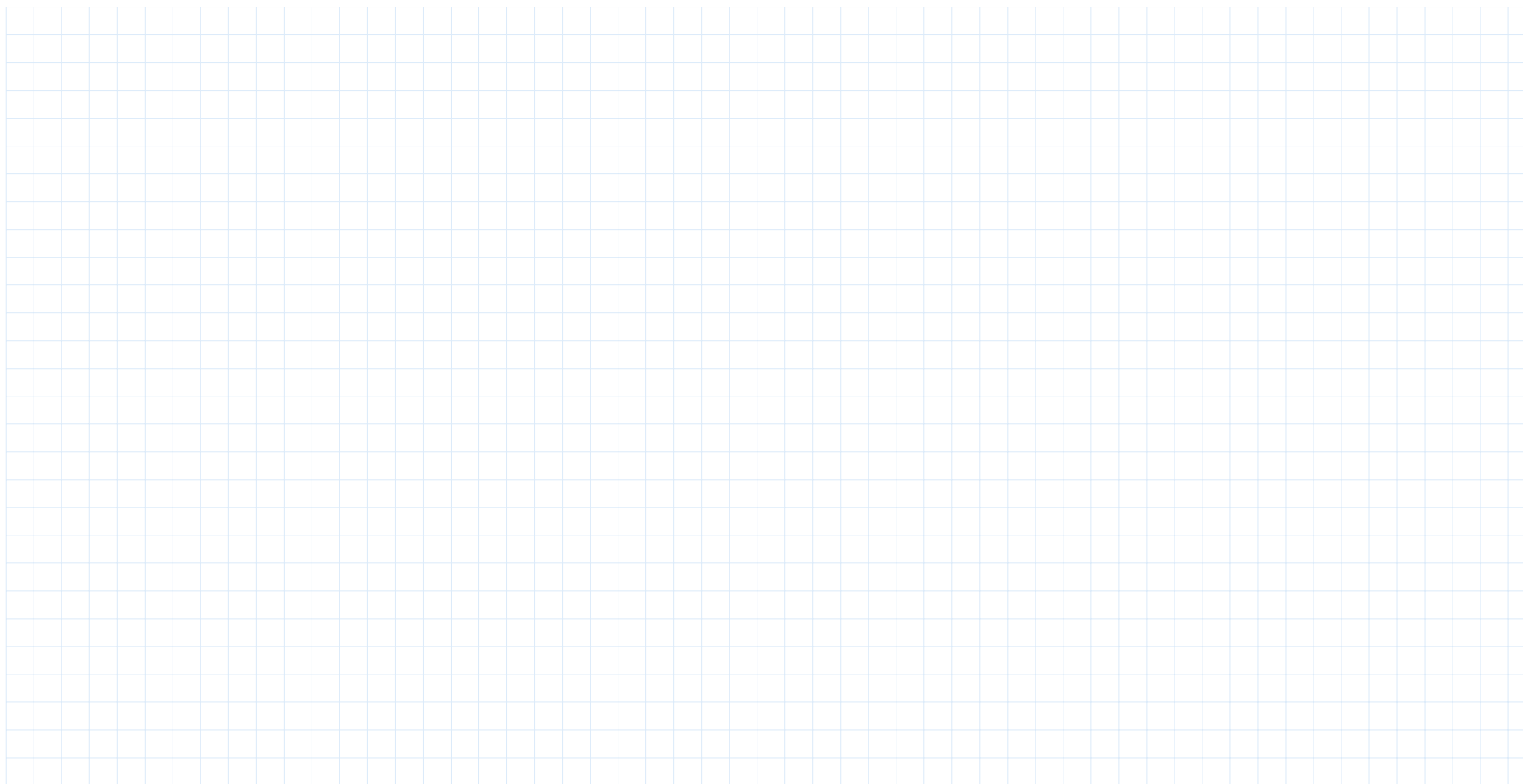


Reduce a Power

Rearranging the cosine double-angle formulas gives

$$\sin^2 x = \frac{1 - \cos(2x)}{2}, \quad \cos^2 x = \frac{1 + \cos(2x)}{2}.$$

Rewrite $\cos^4 x$ with no sine or cosine power greater than 1.

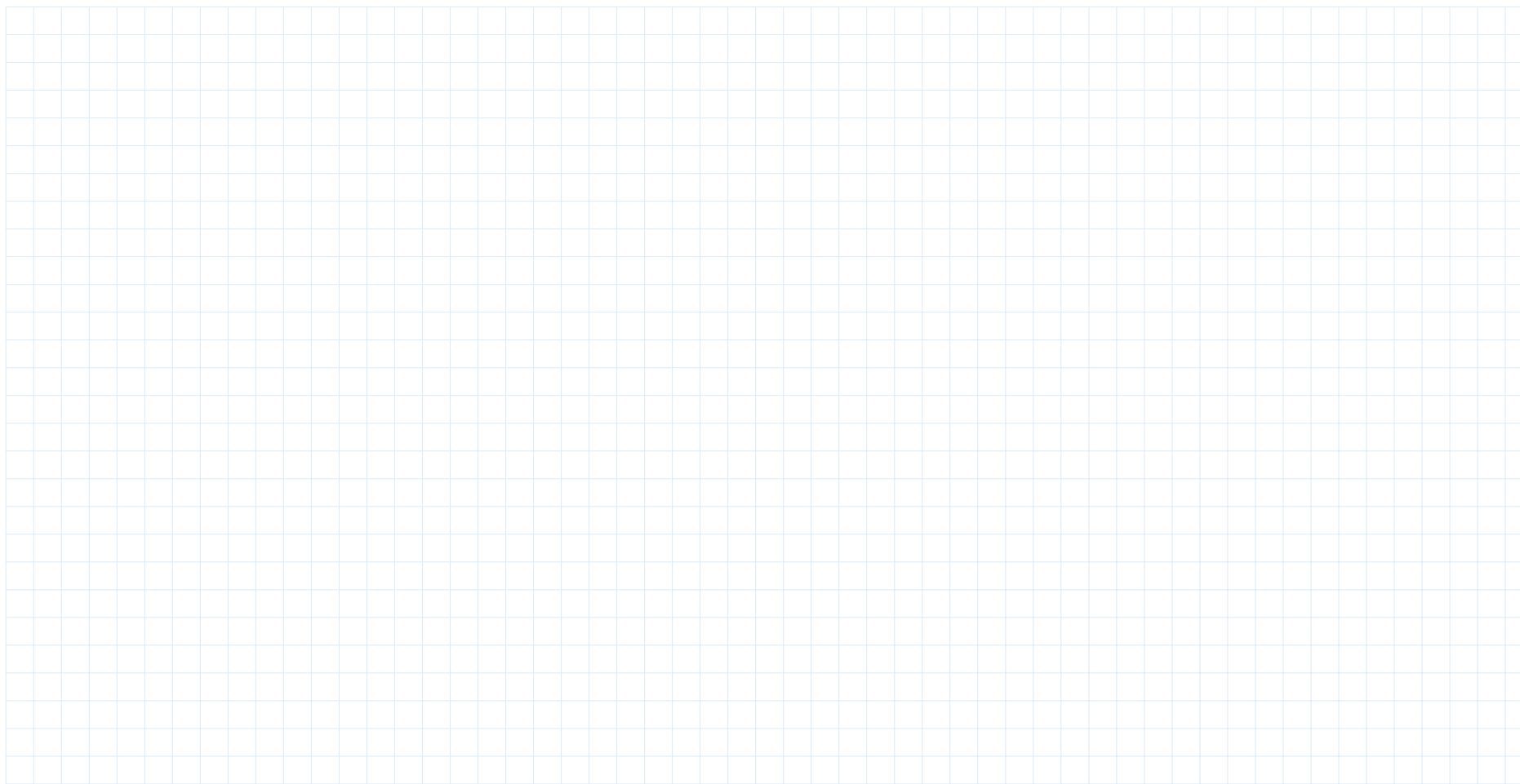


Solve with a Double Angle

Solve on $0 \leq \theta < 2\pi$:

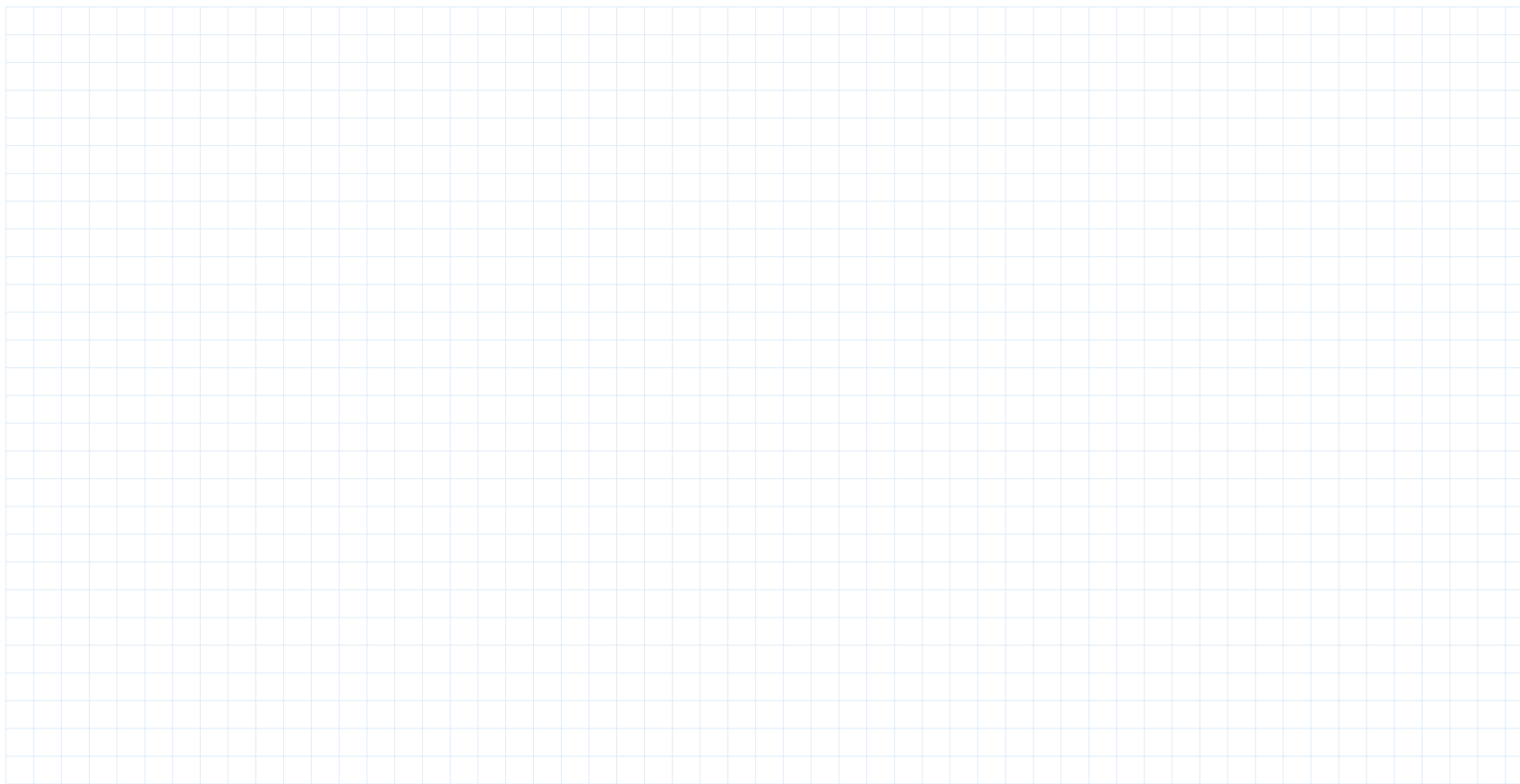
$$\sin \theta \cos \theta = -\frac{\sqrt{3}}{4}.$$

Use a double-angle identity, and keep track of the interval for 2θ .



$$\cos \frac{\alpha}{2} = \pm \sqrt{\frac{1 + \cos \alpha}{2}}.$$

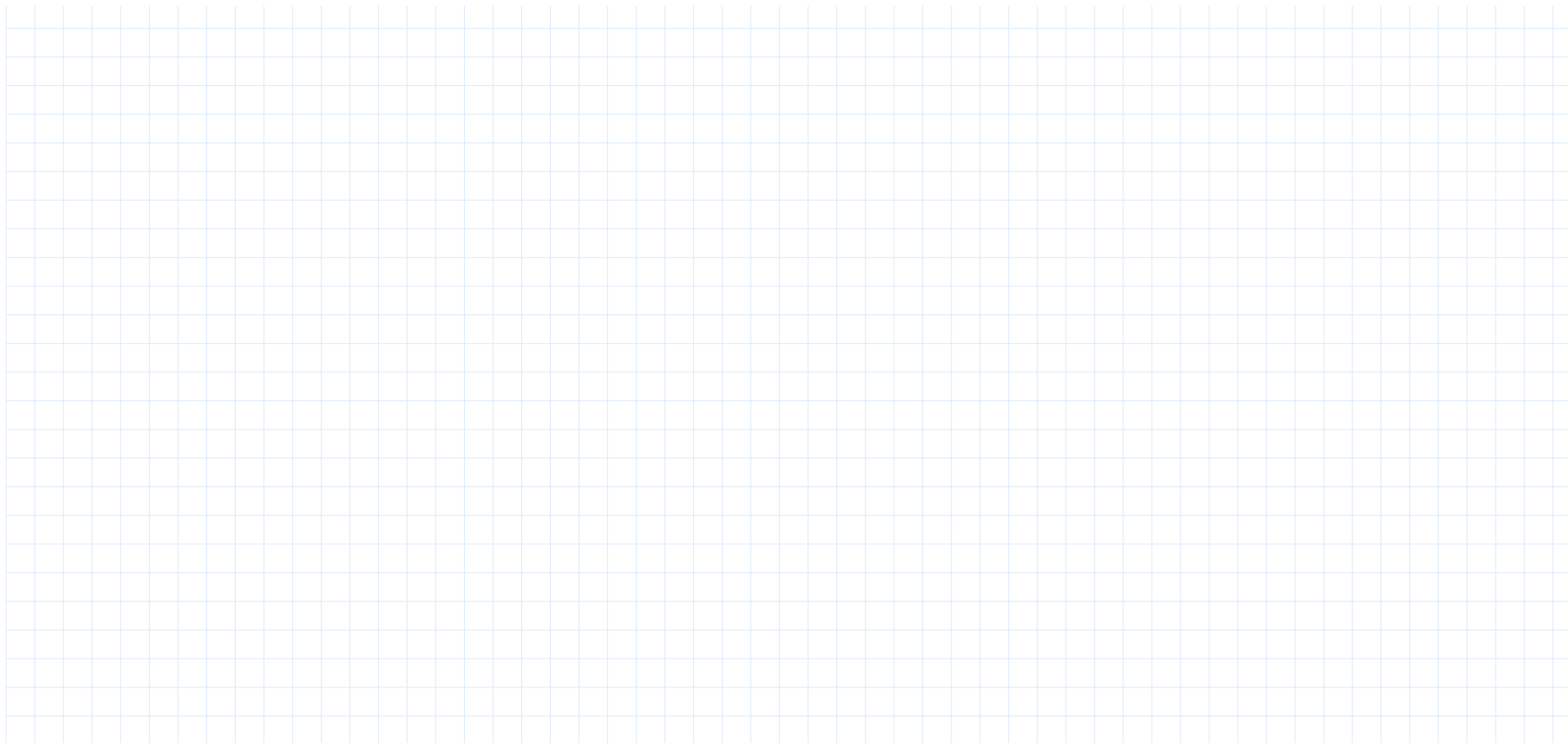
The sign depends on the quadrant of $\frac{\alpha}{2}$. Use $75^\circ = \frac{150^\circ}{2}$ to find $\cos 75^\circ$ exactly.



Choose the Half-Angle Signs

$$\sin \frac{\theta}{2} = \pm \sqrt{\frac{1 - \cos \theta}{2}}, \quad \cos \frac{\theta}{2} = \pm \sqrt{\frac{1 + \cos \theta}{2}}.$$

Suppose $\cos \theta = -\frac{7}{25}$ and $\pi < \theta < \frac{3\pi}{2}$. Find $\sin \frac{\theta}{2}$, $\cos \frac{\theta}{2}$, and $\tan \frac{\theta}{2}$ exactly.



An Application and an Extension

Ignoring air resistance, a projectile launched at 64 feet per second has horizontal range

$$R(\theta) = \frac{64^2}{16} \sin \theta \cos \theta, \quad 0 < \theta < \frac{\pi}{2}.$$

Rewrite R using $\sin(2\theta)$. Find the launch angle and maximum range. **Optional extension.** Use $\sin(2\theta + \theta)$ to derive a formula for $\sin(3\theta)$ in terms of $\sin \theta$ alone.

